

Adverse selection

Asymmetric information

- Asymmetry of information often breaks optimality properties of markets.
Two main phenomena with many variations.
 - Adverse selection (hidden information).
 - Moral hazard (hidden action).

Adverse selection

- An agent's trading decisions depend on her private information. The informed agent's actions then reveal information. This affects, adversely, the uninformed side.
- Ex: Agents have private information about their innate health needs and then choose among insurance policies.
- Common models: signaling, screening, mechanism design in various models.

Moral hazard

- An agent's actions cannot be freely monitored, but affect other agent's utility.
- Example: The success of a firm depends on its manager's effort. Effort is unobservable by the owner. It may not be possible to induce the manager to provide the desired effort level.
- Common model: principal-agent.

Adverse selection

- Topics:
- Adverse selection and market inefficiency.
 - Akerlof's "Market for lemons".
 - Second best outcome: Constrained optimality
 - Signaling models
 - Screening models

The market for lemons

I wrote “The Market for Lemons,” (a 13-page paper for which I was awarded the Prize in Economics) during my first year as assistant professor at Berkeley, in 1966-67. It concerns how horse traders respond to the natural question: “if he wants to sell that horse, do I really want to buy it?” Such questioning is fundamental to the market for horses and used cars, but it is also at least minimally present in every market transaction. — George Akerlof

George Akerlof, Writing the Market for Lemons: A Personal and Interpretive Essay.

[link_to_essay](#)

Market for lemons

- Objective: Asymmetry of information causes markets to fail. Often markets require regulation for efficiency.
- Market for used cars.
- Asymmetry of information: The seller has better information than the buyer about the “quality” of the car.
- See Akerlof (1970).

Example

- Depending on its *quality* a car is worth x in $\{10, 12, 14, 16, 18, 20\}$ where x is expressed in thousands.
- Seller knows quality x of her car but buyer does not.
- Seller only sells if the price $p \geq x$.
- Buyer only buys if $p \leq E(x)$.
- All values of x are equally likely.
- What is the market price p and what type of cars will be traded?

Suppose $p > 20$.

- Sellers wish to sell since $(\forall x) p > 20 \geq x$.
- Buyer does not buy:

$$E(x) = \frac{1}{6}10 + \frac{1}{6}12 + \frac{1}{6}14 + \frac{1}{6}16 + \frac{1}{6}18 + \frac{1}{6}20 = 15$$

- and $p > 20 > 15 = E(x)$.

Suppose $p = 15$.

- Sellers with $x \leq 15$ wish to sell. Sellers with $x > 15$ do not.
- Buyers anticipate that sellers with $x > 15$ do not sell.

$$E(x) = \frac{1}{3}10 + \frac{1}{3}12 + \frac{1}{3}14 = 12$$

Since $p = 15 > 12 = E(x)$, buyers do not buy ($E(x) - p < 0$).

Example

Suppose $p = 10$.

- Sellers of lemons ($x = 10$) wish to sell. Sellers with $x > 10$ do not.
- Buyers anticipate that sellers with $x > 10$ do not sell.

$$E(x) = \frac{1}{1} 10 = 10$$

- Buyers buy lemons; $p = 10 = E(x)$, knowing what they buy.

Model

- The seller observes her type $x \in [x', x'']$; x is the value to the seller.
- The type x distributed according to density $f(x)$.
- The buyer does not observe x .
- The buyer is willing to pay $v(x)$ for car x ; buyer is expected utility maximizer.
- Positive gains from trade: $(\forall x \in (x', x'']) v(x) > x$ and $v(x') = x'$.
- The functions f and v and the structure of the model are common knowledge.

(Both buyer and seller believe x is distributed according to f , common prior assumption.)

Benchmark: Complete information

- x is publicly observed before any transaction is arranged.
- Market outcome:
- Seller sells only if $p \geq x$.
- Buyer buys only if $v(x) \geq p$.
- Provided $v(x) \geq p \geq x$, trade takes place.
- Gains from trade are realized and divided among participants.

Incomplete information

The buyer's expected value of a car depends on the buyer's beliefs about the cars in the market.

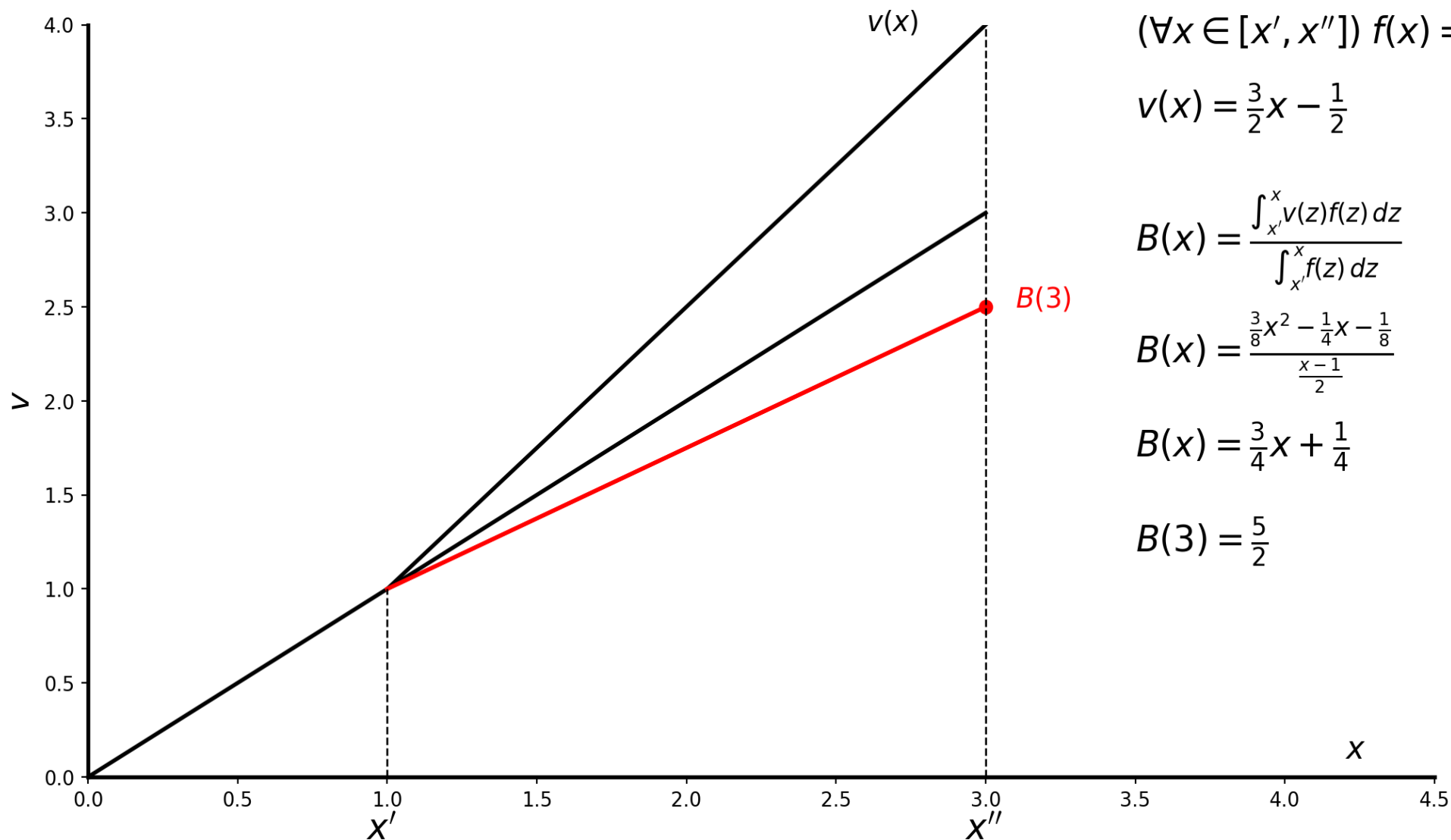
If types in market are $[x', x] \subseteq [x', x'']$, the buyer's expected value of a car is

$$B(x) = \frac{\int_{x'}^x v(z) f(z) dz}{\int_{x'}^x f(z) dz}$$

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- Set price p . Then $x > p \implies [x \text{ is not in the market}]$.
 - The buyer's expected value of the car when price is p

$$B(p) = \frac{\int_{x'}^p v(z) f(z) dz}{\int_{x'}^p f(z) dz}$$

Graph, lemons example



$$(\forall x \in [x', x'']) f(x) = \frac{1}{2}$$

$$v(x) = \frac{3}{2}x - \frac{1}{2}$$

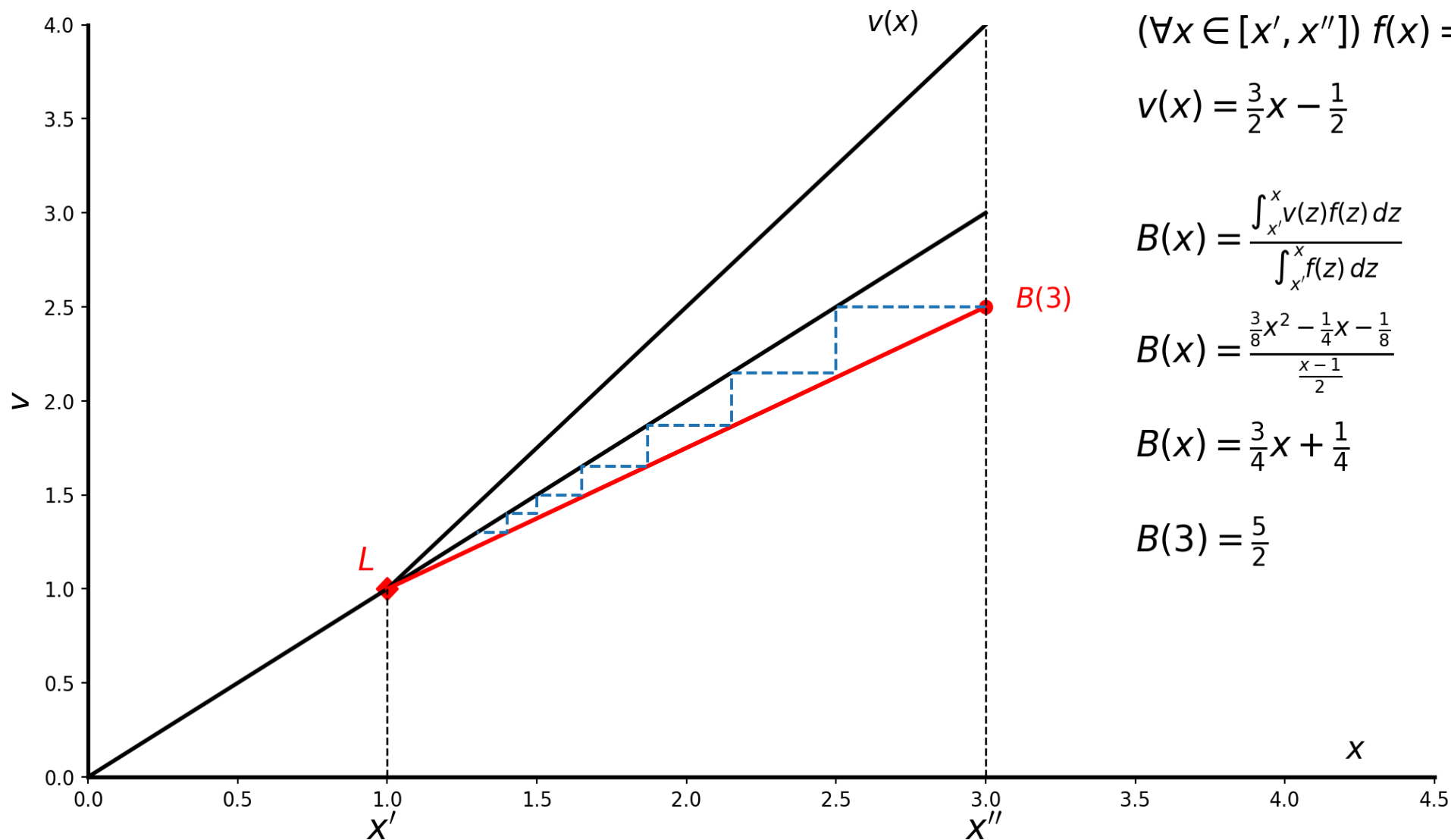
$$B(x) = \frac{\int_{x'}^x v(z) f(z) dz}{\int_{x'}^x f(z) dz}$$

$$B(x) = \frac{\frac{3}{8}x^2 - \frac{1}{4}x - \frac{1}{8}}{\frac{x-1}{2}}$$

$$B(x) = \frac{3}{4}x + \frac{1}{4}$$

$$B(3) = \frac{5}{2}$$

Graph, lemons example 2



$$(\forall x \in [x', x'']) f(x) = \frac{1}{2}$$

$$v(x) = \frac{3}{2}x - \frac{1}{2}$$

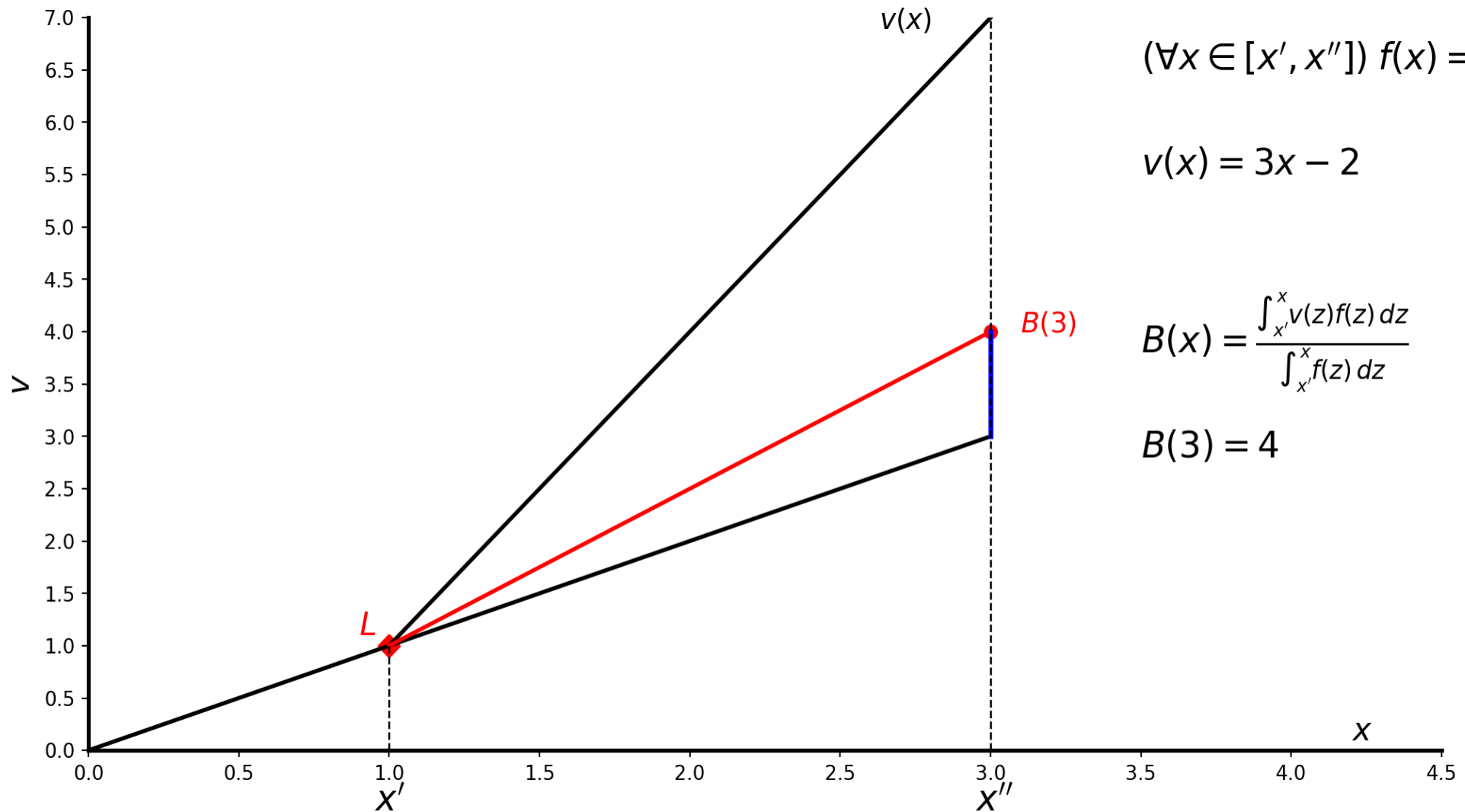
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$$B(x) = \frac{\frac{3}{8}x^2 - \frac{1}{4}x - \frac{1}{8}}{\frac{x-1}{2}}$$

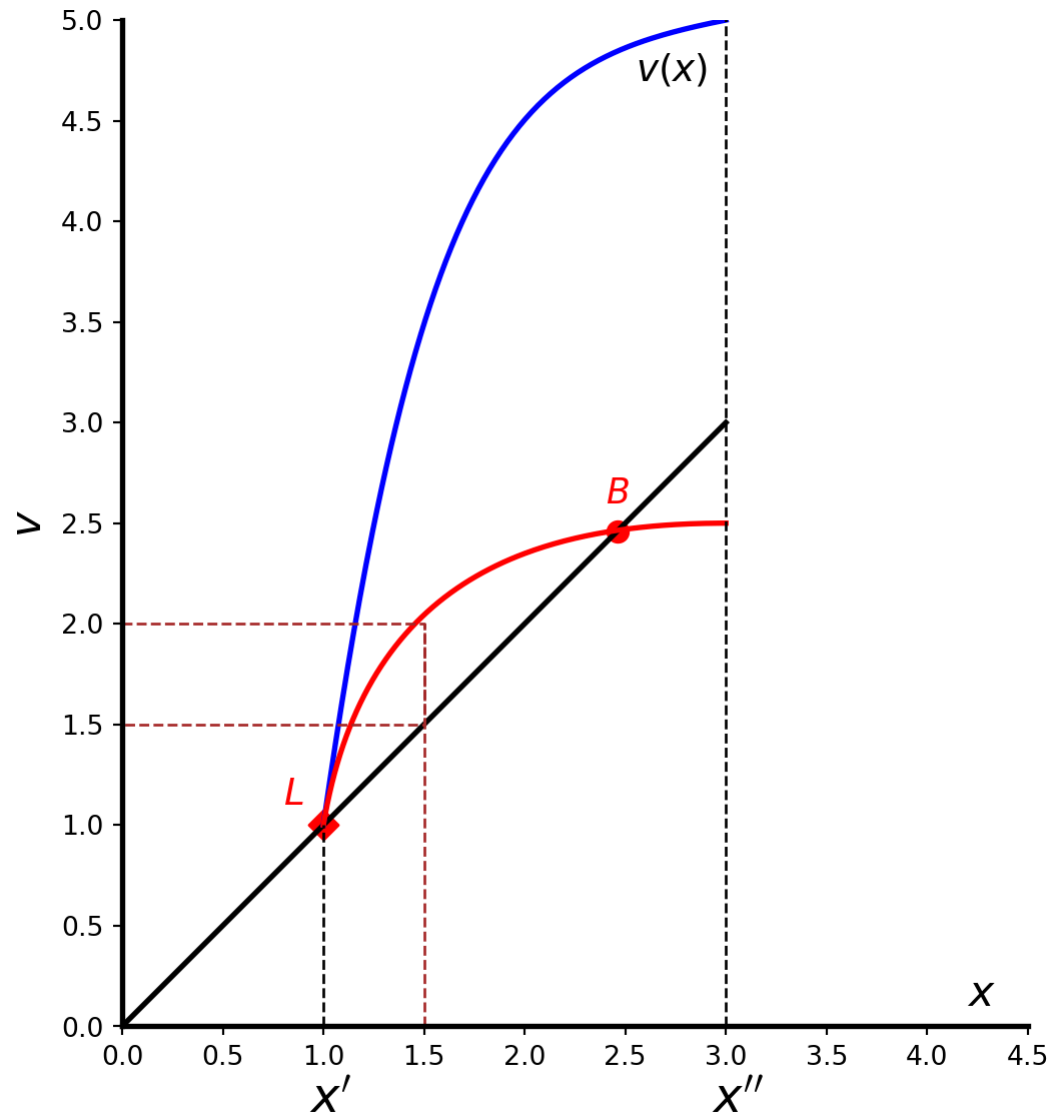
$$B(x) = \frac{3}{4}x + \frac{1}{4}$$

$$B(3) = \frac{5}{2}$$

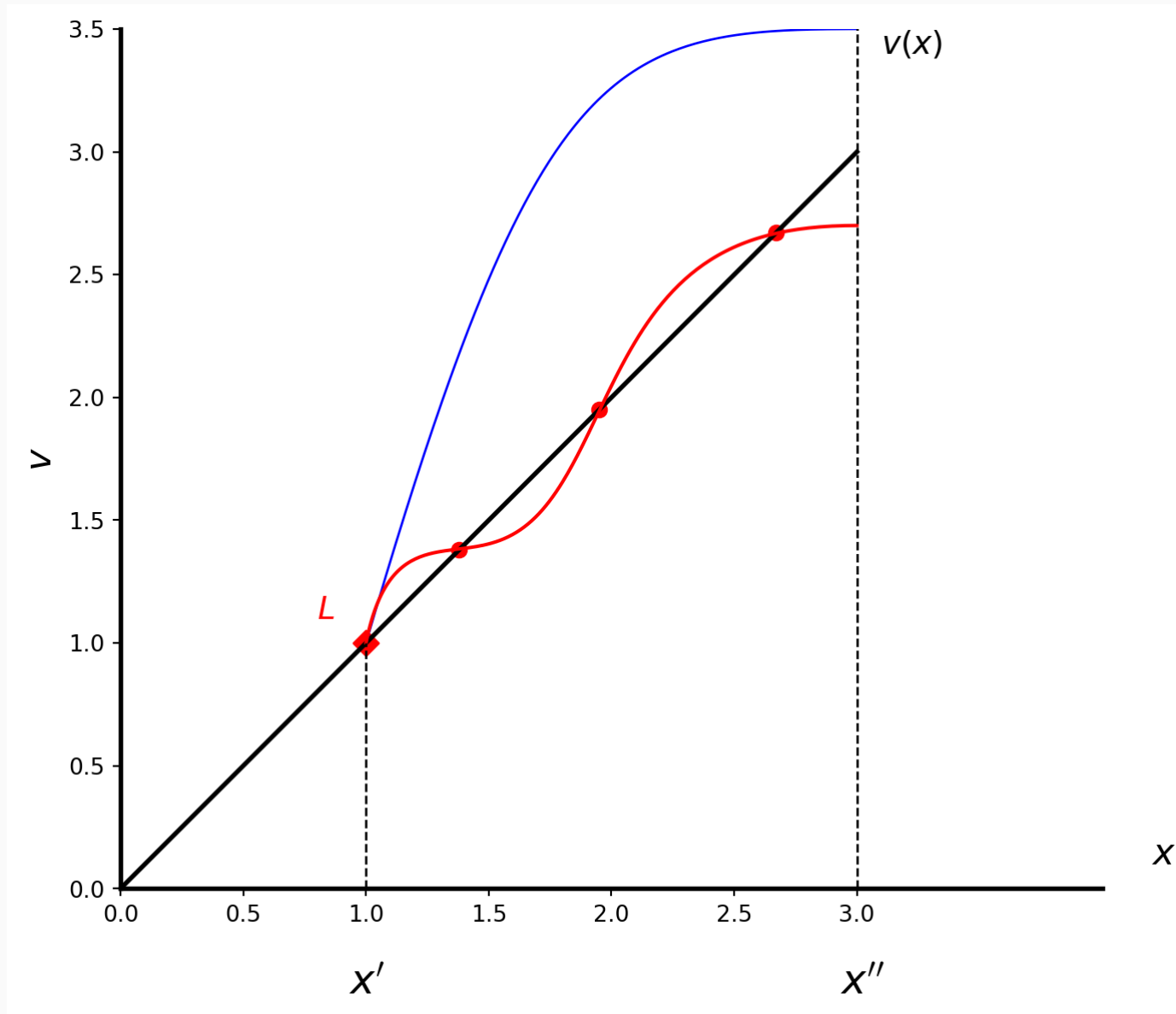
Graph, Not lemons



Other possibilities



Constrained Pareto optimal solution



Responses to adverse selection

- What can the seller do?

The seller, the informed party, can try to signal her information.
(Signaling model)

- What can the buyer, the uninformed party, do?

Could the buyer buy the information he is missing from the seller? How?

Signaling and screening

- Adverse selection leads to inefficiencies. Sometimes these inefficiencies originate on the inability of the uninformed party to separate the different types.

Signaling or screening practices.

- Signaling. The *good* type tries to separate from the *bad* type using signals.
- Screening. The uninformed party proposes contracts, such as multiple product lines, to screen consumers of different types.

Ex. Third degree price discrimination, insurance, etc.

References

Akerlof, George A. 1970. "The Market for "Lemons": Uncertainty and the Market Mechanism." *Quarterly Journal of Economics* 84: 488–500.